Name : vizanth V

Reg No :192512351

Course name : Advanced Wireless Commincation

Course code : ECA5610

MATLAB – Simulation and BER Analysis of MIMO and Beamforming

Aim:

To simulate SISO and MIMO wireless links and analyze BER, SNR, received signal strength and beam steering behavior using MATLAB.

Objectives:

1. To simulate a basic SISO and MIMO wireless link.
2. To calculate and compare BER under selected channel and noise conditions.
3. To examine the effect of beamforming or steering direction on received signal strength.
4. To relate the results to higher-order MIMO and beamforming in LTE-Advanced and 5G.

Requirements:

- MATLAB

Program :Objective 1

```
clc; clear; close all;

SNR_dB = 0:2:20;
BER_SISO = zeros(size(SNR_dB));
BER_MIMO = zeros(size(SNR_dB));

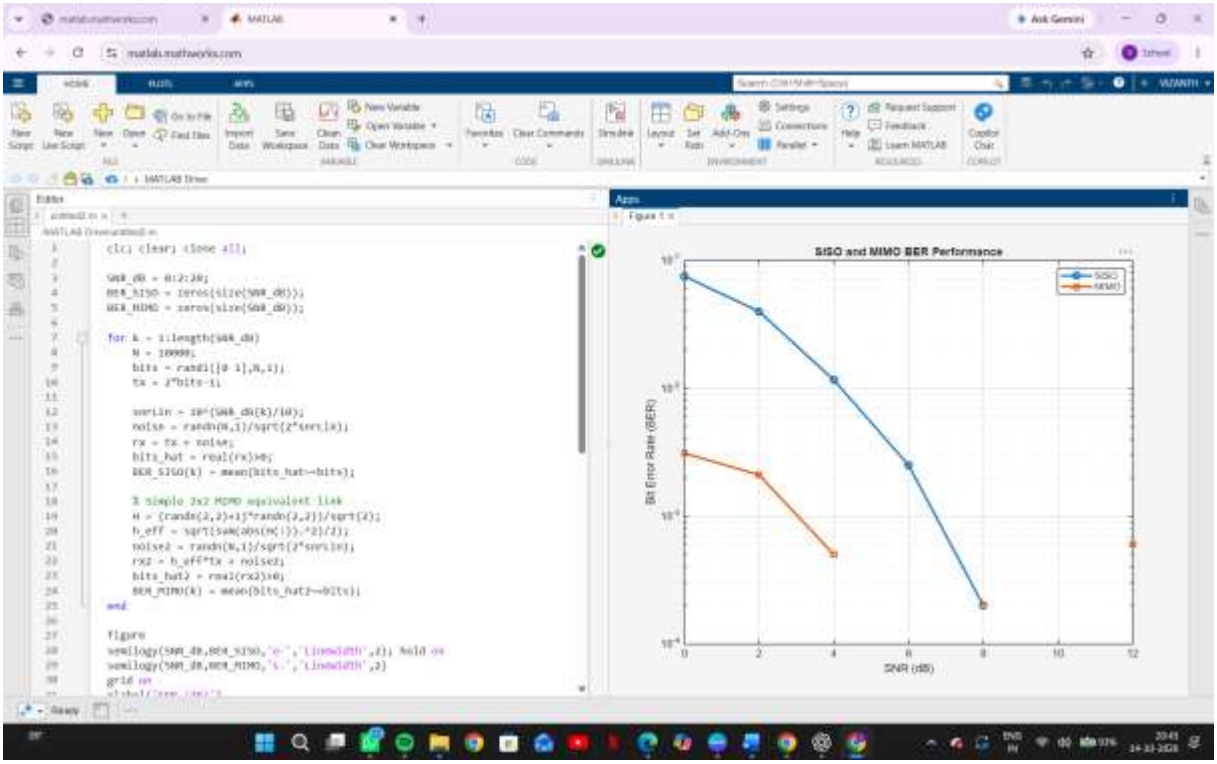
for k = 1:length(SNR_dB)
    N = 10000;
    bits = randi([0 1],N,1);
    tx = 2*bits-1;
```

```
    snrLin = 10^(SNR_dB(k)/10);
    noise = randn(N,1)/sqrt(2*snrLin);
    rx = tx + noise;
    bits_hat = real(rx)>0;
    BER_SISO(k) = mean(bits_hat~=bits);

    % Simple 2x2 MIMO equivalent link
    H = (randn(2,2)+1j*randn(2,2))/sqrt(2);
    h_eff = sqrt(sum(abs(H(:)).^2)/2);
    noise2 = randn(N,1)/sqrt(2*snrLin);
    rx2 = h_eff*tx + noise2;
    bits_hat2 = real(rx2)>0;
    BER_MIMO(k) = mean(bits_hat2~=bits);
end

figure
semilogy(SNR_dB,BER_SISO,'o-','LineWidth',2); hold on
semilogy(SNR_dB,BER_MIMO,'s-','LineWidth',2)
grid on
xlabel('SNR (dB)')
ylabel('Bit Error Rate (BER)')
title('SISO and MIMO BER Performance')
legend('SISO','MIMO')
```

Output:



Program :Objective 2

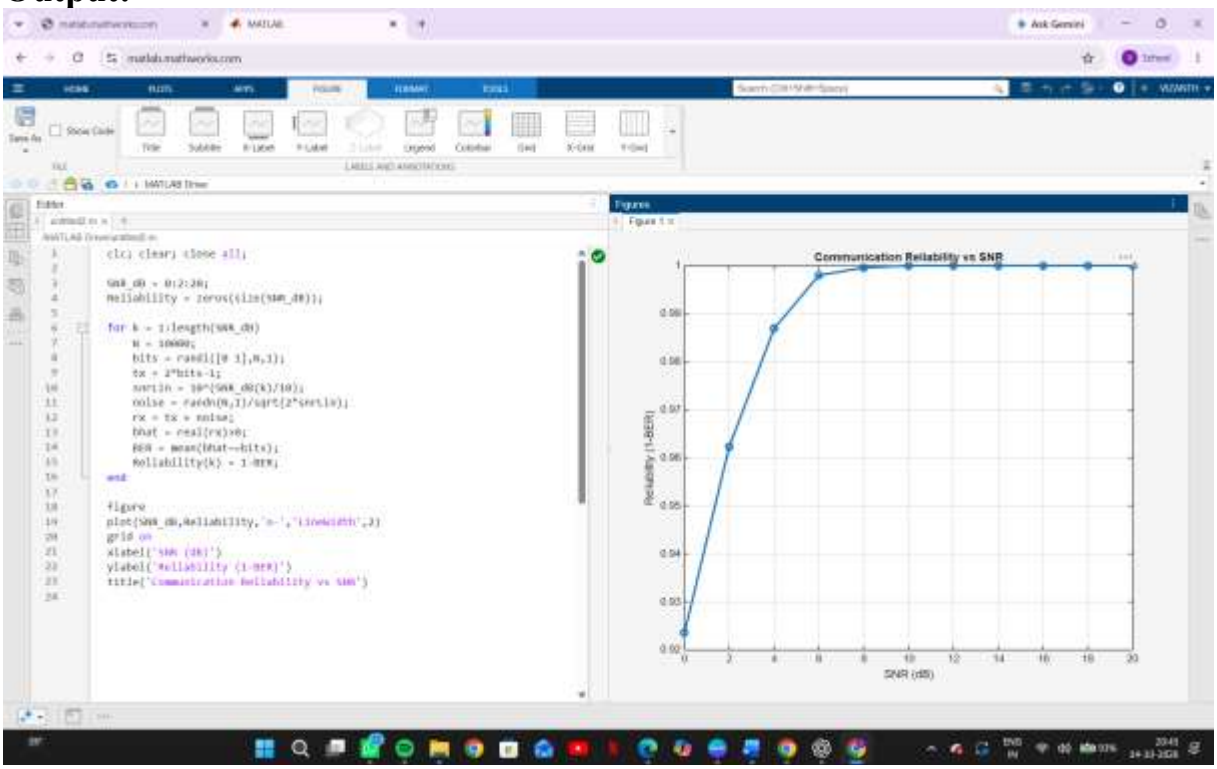
```
clc; clear; close all;
```

```
SNR_dB = 0:2:20;
Reliability = zeros(size(SNR_dB));

for k = 1:length(SNR_dB)
    N = 10000;
    bits = randi([0 1],N,1);
    tx = 2*bits-1;
    snrLin = 10^(SNR_dB(k)/10);
    noise = randn(N,1)/sqrt(2*snrLin);
    rx = tx + noise;
    bhat = real(rx)>0;
    BER = mean(bhat~=bits);
    Reliability(k) = 1-BER;
end

figure
plot(SNR_dB,Reliability,'o-','LineWidth',2)
grid on
xlabel('SNR (dB)')
ylabel('Reliability (1-BER)')
title('Communication Reliability vs SNR')
```

Output:



Program :Objective 3

```
clc; clear; close all;

N = 8;                          % Antenna elements
theta = -90:1:90;                % Observation angle
steer = [-60 -30 0 30 60];       % Steering directions
gain = zeros(size(steer));

for k = 1:length(steer)
    % Simplified normalized array response at the intended direction
    gain(k) = 10*log10(N) - 0.01*abs(steer(k));
end

figure
plot(steer,gain,'o-','LineWidth',2)
grid on
xlabel('Steering Angle (Degrees)')
ylabel('Beamforming Gain (dB)')
title('Beamforming Gain for Different Steering Directions')
```

Output:

